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Section : AIML (section-2)

MFML Assignment 1

A1.) Given the matrix L is a lower triangular, we can use forward substitution to solve the system $Lx=b$. Forward substitution works by solving the system row by row from top to bottom.

Here, an additional constraint that only a single row of the matrix can be stored in memory at a time. Hence, the constraint isn't preventing us from using forward substitution, we will also minimise memory writes by only updating our solution vector x .

Here is an algo description that solves this system with these constraints:

Step 1: Initialise x as a non-zero vector with length 100000

Step 2: For i from 1 to 100000:

Step 3: Load row i of L into memory

Step 4: Let $sum = 0$

Step 5: For j from 1 to $i-1$:

Step 6: $sum += L[i][j] * x[j]$

Step 7: $x[i] = (b[i] - sum) / L[i][i]$

Only load one row of L into memory at a time, thus satisfying the first constraint. We also only write into the solution vector x in memory, & this write is necessary for the solution, thus minimising the no. of writes to memory.

- A-2> 1. It is an elimination matrix - lower triangle
2. Lower triangle matrix & its property

E_{ij} will be

$$\begin{bmatrix} e_{11} & 0 & 0 & \dots \\ 0 & e_{22} & 0 & \dots \\ 0 & 0 & e_{33} & \dots \\ 0 & 0 & \dots & \textcircled{\alpha} & e_{55} \end{bmatrix}$$

$\xrightarrow{\quad} 5^{\text{th}} \text{ row \& 2}^{\text{nd}} \text{ column}$
 e_{52} is non-zero

E_{pq} will be

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & \dots \\ 0 & 1 & 0 & 0 & 0 & \dots \\ 0 & 0 & 1 & 0 & 0 & \dots \\ 0 & 0 & 0 & 1 & 0 & \dots \\ \textcircled{\beta} & 0 & 0 & 0 & 1 & \dots \end{bmatrix}$$

$\xrightarrow{\quad} e_{pq}$

When we multiply both of the above i.e.,

$$\begin{aligned} &E_{ij} \times E_{pq} \\ &\& E_{pq} \times E_{ij} \end{aligned}$$

And put equal we will get an equation in terms of α and β which gives us the required condition & hence the set of elimination matrix E_{pq}

commutativity means $AB = BA$ (Matrix multiplication)

$\therefore$ If $p \neq i$ and $q \neq j$,
then $E_{pq} E_{ij} = E_{ij} E_{pq}$, and
the elimination matrices commute. [Example py program
cnld...]

A3.7 Now, given condition is
 Eigen value remains complex for $0 \leq n \leq 49$
 and for A^{50} remains real.

Now, divide 2π in 50 parts, so we get,

$$\theta = \frac{2\pi}{50} \text{ for satisfying condition } \lambda = 1 \text{ with modulus 1 \& being a complex number.}$$

$$\text{Now, let } \theta = \frac{2\pi}{50}$$

$$\Rightarrow A = \begin{bmatrix} \cos 2\pi/50 & -\sin 2\pi/50 \\ \sin 2\pi/50 & \cos 2\pi/50 \end{bmatrix}$$

Now, applying De Moivre's theorem,

$$z = r(\cos \theta + i \sin \theta)$$

$$\Rightarrow z^n = r^n (\cos n\theta + i \sin n\theta)$$

$$\text{we get, } A^{50} = \begin{bmatrix} \cos 2\pi/50 \times 50 & -\sin 2\pi/50 \times 50 \\ \sin 2\pi/50 \times 50 & \cos 2\pi/50 \times 50 \end{bmatrix}$$

$$\Rightarrow A^{50} = \begin{bmatrix} \cos 2\pi & -\sin 2\pi \\ \sin 2\pi & \cos 2\pi \end{bmatrix}$$

which is an identity matrix ($\because \cos^2 \theta + \sin^2 \theta = 1$)

$$\Rightarrow A^{50} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow$ Eigen values of A^{50}

$$\Rightarrow \begin{bmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{bmatrix} = 0 \quad (\because (1-\lambda)^2 = 0)$$

$$\Rightarrow \lambda = 1, -1 \in \mathbb{R}$$

So, for set of $\forall A = \{A^1 \text{ to } A^{49}\}$

$$A^1 \text{ to } A^{49} = \begin{bmatrix} p & -s \\ q & s \end{bmatrix}$$

When $p = s = \cos \frac{2\pi}{50}$ (1 to 49) i.e., $< \cos 2\pi < 1$

$$s = -s = \sin \frac{2\pi}{50} \text{ (1 to 49) i.e., } < \sin 2\pi \text{ (1 to 49)} < \cos 2\pi/50$$

$\Rightarrow$ determinant $\begin{bmatrix} p & -s \\ q & s \end{bmatrix}$ is always complex.

$$\Rightarrow \text{resultant matrix} = \begin{bmatrix} \cos 2\pi/50 & -\sin 2\pi/50 \\ \sin 2\pi/50 & \cos 2\pi/50 \end{bmatrix}$$

This satisfies both conditions

Modulus = 1

$\Delta \lambda = \text{complex number}$

A.4 for a simple triangular matrix upper/lower
Eigen values are values in diagonal itself

$$\begin{bmatrix} c_{11} & 0 & 0 & 0 \\ .. & c_{22} & 0 & 0 \\ .. & .. & c_{33} & 0 \end{bmatrix} \rightarrow \begin{bmatrix} (c_{11} - \lambda) & 0 & 0 & 0 \\ 0 & (c_{22} - \lambda) & 0 & 0 \\ e & d & e & .. \end{bmatrix}$$

Hence,

$$(c_{11} - \lambda)(c_{22} - \lambda)(c_{33} - \lambda) \dots = 0$$

$$\lambda = \{c_{11}, c_{22}, .. c_{33} \dots c_{nn}\}$$

So, B can query only diagonal element to get the answer.

$\rightarrow$ for calculating of Eigen value

A-5) $C \in \mathbb{R}^{n \times n}$ $C^T = C$

$\therefore C$ is symmetric

i.e., $a_{ij} = b_{ji}$

Let's take an example of a symmetric matrix

Eg:
$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Eigen vectors of symmetric matrix and Eigen values of sym. matrix are corresponding to, orthogonal.

Orthogonal $n \times n$

The columns or rows are orthogonal i.e., dot product = 0

If A is orthogonal $\therefore AA^{-1} = I$

$\Rightarrow A^T \cdot A = I$ and $A^{-1} = A^T$

Euclidean norm & Symmetric Matrix

Magnitude of a vector

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

$\Rightarrow \|x\|_2 > 0$ if $x \neq \phi$

$\Rightarrow \|\alpha x\|_2 = |\alpha| \|x\|_2$ where α = scalar length of sum

$\|x+y\|_2 \leq \|x\|_2 + \|y\|_2$

of 2 vectors is $\leq$ length of sum of individual lengths
triangle inequality

$CV = \lambda V$

where V = Orthogonal

$O^T O = I$

Diagonalization of A only $n \times n$

$$A = PDP^{-1}$$

where, D = diagonal matrix with eigen values of A
 P = columns of eigen vectors

$$\rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad ac + bd = 0$$

$$S \rightarrow D$$

$$A = U \Sigma V^T$$

$$\Rightarrow A^T \cdot A \Rightarrow |A^T \cdot A - \lambda I| = 0$$

from λ we get σ

$$\sigma_1 = \sqrt{\lambda_1}, \sigma_2 = \sqrt{\lambda_2}, \dots$$

$$\therefore \Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$$

Left Singular Vector (U)

$$A^T \cdot A \cdot U_1 = \lambda_1 \cdot U_1 \quad \text{where } [U_1, U_2, U_3] = U$$

Right Singular Vector (V)

$$A \cdot A^T \cdot V_1 = \lambda_1 \cdot V_1$$

$$\Rightarrow [V_1, V_2, V_3] = V^T$$

$$\text{Now,} \quad A = U \Sigma V^T \quad [\because \text{SVD of } A]$$

where, $U, V \rightarrow$ orthogonal matrices

$\Sigma \rightarrow$ diagonal matrix

$$\therefore V \Sigma^T \Sigma V^T = U \Sigma \Sigma^T U^T$$

$$\Rightarrow C^T C = C C^T \Rightarrow \boxed{U = V}$$

